

Carlo Vilorio

Software & AI Developer | IT Support Specialist

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SUMMARY

Computer Science graduate (UH Mānoa, 2024) with a continuous track record of building and deploying AI-powered and full-stack applications, plus hands-on Windows system administration — most recently a Windows Server 2022 Active Directory lab with Group Policy and PowerShell onboarding automation. Experienced diagnosing hardware, software, and network issues in university IT environments. Fluent communicator who adapts technical concepts for any audience — actively seeking a service desk or software support role at a Hawaii-based tech company.

TECHNICAL SKILLS

Languages: Python, Java, C#, JavaScript, HTML/CSS, PowerShell

Frameworks & DBs: React, Node.js, ASP.NET Core, Entity Framework Core, PostgreSQL, MySQL, SQL Server

AI / ML: OpenAI GPT API, scikit-learn, Jupyter Notebooks, LLM prompt engineering

Systems & Networking: Windows Server 2022, Active Directory, Group Policy, VirtualBox, Linux/Unix, TCP/IP, DNS, DHCP

Tools & Platforms: Git/GitHub, Docker, Vercel, Render, VS Code, IntelliJ, Figma

IT Support: Hardware/software/network troubleshooting, ticketing systems, knowledge-base documentation, security best practices

Other: Agile/Scrum • CompTIA A+ (in progress) • DoD Secret clearance eligible • English, Tagalog, Korean

EXPERIENCE & PROJECTS

Active Directory Home Lab — Windows System Administration

Jun – Jul 2026

Personal lab • github.com/carloviloria808/ad-home-lab

- Built and administer a Windows domain in VirtualBox: promoted a Windows Server 2022 domain controller (AD DS + DNS, static IP) hosting lab.local, with organizational units, user accounts, and least-privilege security groups.
- Enforced security via Group Policy — domain password policy and role-based desktop lockdown — verified end-to-end on a domain-joined Windows 11 Pro client (gpupdate/gpresult).
- Automated user onboarding in PowerShell: bulk-created 20 accounts from CSV via New-ADUser with temp-password and change-at-first-logon flow.
- Diagnosed, resolved, and documented four real incidents from the command line — including a client DNS misconfiguration and a password-policy failure cascade — each logged symptom → root cause → resolution (full write-up on GitHub).

TFTools — Full-Stack Web Application

Jan 2025 – Present

Personal project • tft-tools.vercel.app • github.com/carloviloria808/TFTools

- Built and deployed a live full-stack web application using C#, ASP.NET Core, and React — frontend on Vercel, containerized backend (Docker) on Render — applying industry-standard separation of concerns across Models, Controllers, and Data Context layers.
- Migrated the production database from SQL Server to PostgreSQL with Entity Framework Core, converting the schema and seeding data from exported JSON.
- Built a RESTful API handling complex data relationships — nested objects, junction tables, self-referencing foreign keys, and circular-reference serialization.
- Documented and tested all API endpoints via Scalar (OpenAPI); configured CORS policy for secure React-to-backend communication.

MyAIRealtor — Full-Stack AI Web Application

Jun 2024 – Dec 2024

Independent project

- Developed an end-to-end AI-powered real estate platform using React, Node.js, and the OpenAI GPT API to provide personalized property recommendations.
- Implemented a data-analysis pipeline that ingests MLS-style listing data, runs statistical comparisons, and surfaces insights via a conversational chat interface.
- Reduced average property search time in user testing by surfacing ranked results through natural-language queries instead of manual filter navigation.

Academia Arena — Web & Backend Developer

Jan 2024 – May 2024

Independent project

- Designed and built a gamified educational platform using JavaScript, HTML/CSS, and a Node.js backend, enabling students to compete in knowledge-based challenges.
- Integrated a MySQL database to track user progress, scores, and leaderboards, supporting multi-user sessions with live score updates.
- Applied UX research principles (Figma wireframes → functional prototype) and Agile/Scrum practice — sprint planning, retrospectives, Kanban — across the 4-month project.

IT Support Assistant

Aug 2022 – Dec 2023

Information Technology Services — University of Hawai‘i at Mānoa

- Handled 30+ weekly phone and email support tickets, resolving hardware, software, and network issues for students and faculty across campus environments.
- Authored and maintained a troubleshooting knowledge base that improved first-contact resolution rate and reduced repeat ticket submissions.
- Enforced security best practices — OS patching, account management, endpoint compliance — across student lab and administrative systems.
- Delivered clear resolution steps and proactive ticket status updates, maintaining high satisfaction scores in follow-up surveys.

Teaching Assistant — Java / Object-Oriented Programming

Jan 2021 – May 2022

Department of Information & Computer Sciences — UH Mānoa

- Supported 3 lab sections per week, guiding 60+ students through debugging, compiling, and testing Java programs in Linux VM environments.
- Explained OOP concepts (inheritance, polymorphism, data structures) using plain-language analogies, improving student comprehension scores on lab assessments.
- Provided one-on-one office-hour support, adapting communication style to each student's background — a direct parallel to service-desk support expectations.

EDUCATION

Bachelor of Science, Computer Science — University of Hawai‘i at Mānoa

Dec 2024

Associate of Arts, Liberal Arts — Honolulu Community College

May 2018

ADDITIONAL

Certifications: CompTIA A+ (in progress) | Clubs: UH Mānoa Grey Hats (Cybersecurity), Game Dev Club, Esports Club | Security: Eligible for DoD Secret clearance | Languages: English, Tagalog, Korean